CSWP: march on in time

Alichan Borokov, Sam Hemati, Joran Van Haelst

December 2025

1 Summary of the problem

in this assignment we implement the march-on in time algorithm on PEC scatterers. We implemented on TM case of the wave equation. This means that the electric field only contains a z-component and the magnetic field only in the x and y- direction. the problem is also consistent in the z-direction this means the material is symmetric under the z-axis. This reduces our problem to a 2 dimensional problem.

After that we use the greens function in its retarded form to solve the boundary integral equation which has as the unknown the current. At last we discretize the integral equation as done in the assignment. And we us those formulae to describe the different problems.

2 assignment

we validate our code by calculating the time domain scattering at a cylinder.

The formulae description is done in the assignment. Our task is implement the march-on-in-time algorithm on this example and look if this correspondds with the analytical solution.

for the virst part we calculate the timedomain scattering at a cylinder. Using an incoming wave going to the positive x direction.